

CLOSED-LOOP AGRICULTURAL CLIMATE CONTROLLER PROJECT REPORT

25MTRN15I – MICROPROCESSOR SYSTEMS

SUBMITTED BY:

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Objective

The goal of this project is to design and implement a closed loop environment automation controller using PIC16F877A microcontroller. The system is continuously receiving analog data from four environmental nodes: soil moisture, ambient humidity, temperature and light intensity (LDR), which enable the creation of an autonomous regulation for a physical greenhouse ecosystem.

Automated irrigation (via a water pump relay), dynamic climate ventilation (via PWM fan control), and day/night cycle tracking (via PWM grow lights) is managed by an embedded firmware using a cooperative multitasking scheduler with Timer1 hardware interrupts. In addition, the building allows for system reliability through non-volatile EEPROM threshold storage, dry run pump safety algorithm, and a latched Emergency Stop (E-STOP) hardware interlock.

Circuit Schematic

The system hardware is based on the PIC16F877A at 4MHz. Peripheral mapping is spread out over the microcontroller pins to keep high voltage control away from analog signal acquisition:

Configured through ADCON1 for Left-Justified 8-bit conversions,

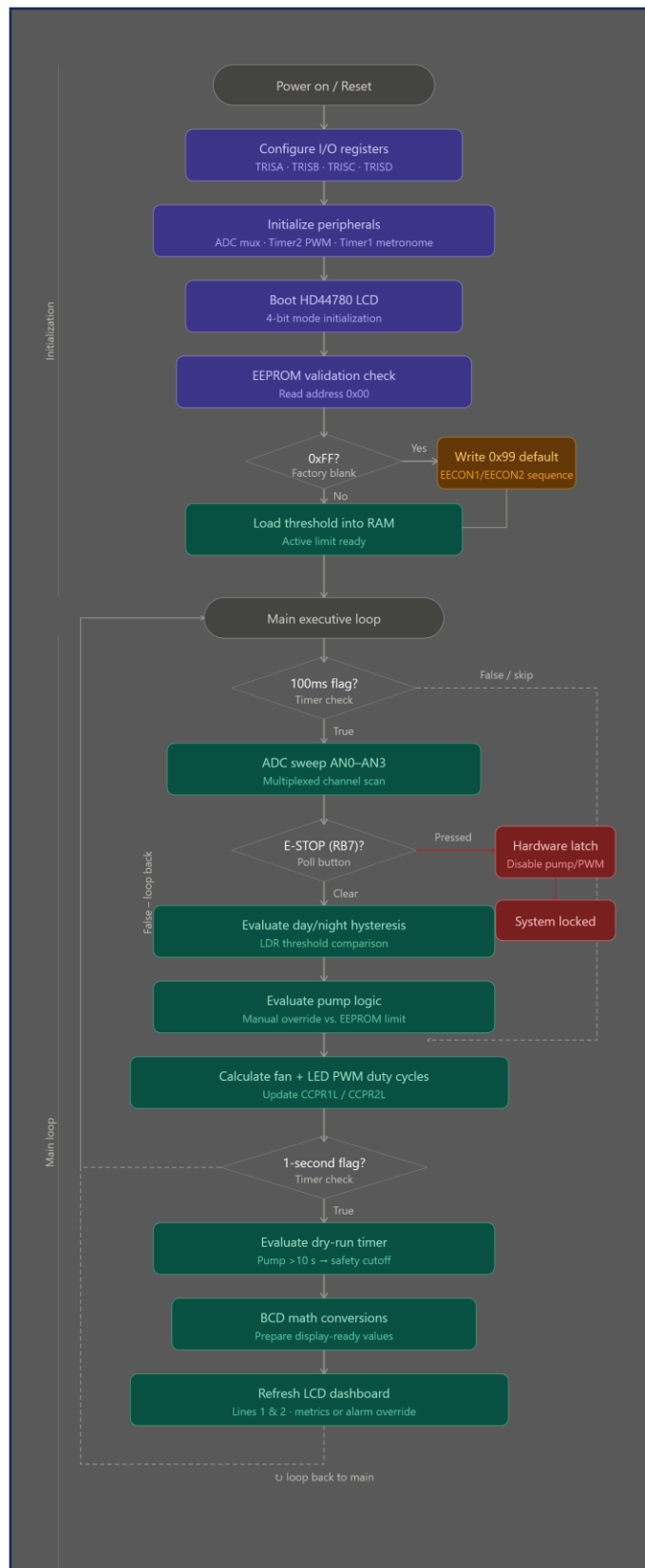
- Port A (Analog Inputs) will contain the Moisture (AN0), Humidity (AN1), Temperature (AN2), and LDR (AN3) sensor circuits.
- Port B (Digital Inputs): Contains active-low manual override switches on the pump, LED and Fan as well as the hardware E-STOP button associated with RB7.
- Port C (Actuator Outputs): Controls heavy load drivers. RC0 controls the pump relay transistor, and RC1 (CCP2) and RC2 (CCP1) provide variable hardware PWM signals to the LED array and ventilation fan.
- Port D (Display Bus): Real time telemetry interface with the HD44780 16x2 Text LCD (4-bit, D0-D3 for data, D4 for RS, D5 for EN).

Main Executive Loop

- Check 100ms Flag If True:
 - Execute multiplexed ADC sweeps across AN0 to AN3.
 - Poll E-STOP button (RB7). If pressed Trigger Hardware Latch (Disable pump/PWM, lock system).
 - Evaluate Day/Night Hysteresis thresholds using LDR data.
 - Evaluate Pump Logic (Manual Override vs. Automated EEPROM limit comparison).
 - Calculate proportional Fan and LED PWM duty cycles and update CCPR1L/CCPR2L.

Check 1-Second Flag If True:

- Evaluate Dry-Run timer (If pump runs >10 seconds, trigger safety cutoff).
- Execute BCD math conversions.
- Refresh LCD Dashboard (Lines 1 and 2) with real-time sensor metrics or Alarm/E-STOP overrides.

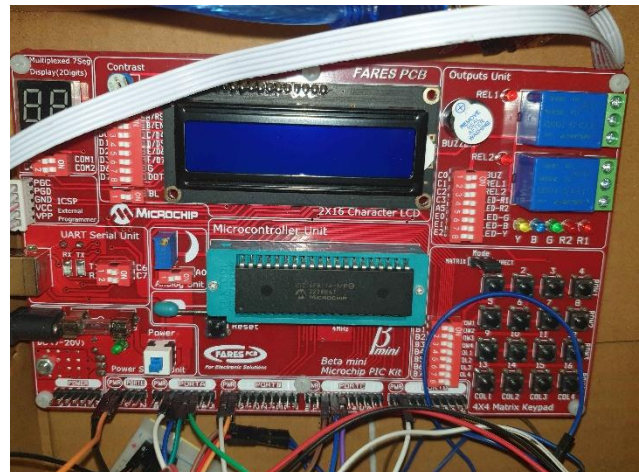


Hardware Implementation

The physical prototype converts the simulated Proteus logic into a real prototype of a breadboard/PCB setup. Appropriate base resistors and switching transistors are used to isolate the high current actuators from the PIC16F877A digital pins, so that reverse current spikes from the pump motor coil do not occur.

Critical System Wiring Components:

- **Flyback Diodes:** A flyback voltage diode was connected on the pump power in case of any back emf to prevent frying circuit components.
- **L298N Module:** The L298N motor driver module offers back-emf, overtemperature protection, controller isolation all in one module hence why it was the perfect choice for peace of mind and it can also be used to power 2 motors(Fan and LED).
- **Power Rail Distribution:** Regulated 5V distributed to the microcontroller, LCD logic, and sensor VCC rails.



Testing & Results

The system has been extensively physically unit-tested to validate the assembly logic with actual electrical parameters. All operational thresholds were passed by the firmware:

1. Analog Multiplexing Stability: The internal ADC was able to perform the analog multiplexing without electrical crosstalk between all 4 analog channels within 100ms. The values of the sensors were updated in the correct RAM registers.
2. When using the LM35 temperature sensor and applying a heat source, the calculated math variable increased dynamically, which was visible as the speed of the motor of the ventilation fan increased (CCP1). By covering the LDR, it turned the system into Night Mode and achieved a duty cycle of 100% (CCP2) for the LED array.
3. Closed-Loop Irrigation: The analog voltage dropped below the EEPROM threshold value (0x99) when the moisture probe was submerged. The Carry-flag logic for the SUBWF was correct, and it was able to immediately ground RC0 to shut off the water pump if it determined there was a wet state.
4. Hardware Safety Interlocks (HSI) Dry Run Alarm function worked as intended: When the sensor was dry, and the pump was already on, the 10-second background timer was triggered and the pump was shut down, with the lcd printout being "ERR:DRY RUN!".

The PWM registers were set to 0x00, the pump relay was switched to 0V and the display was also locked in "E-STOP!" when the E-STOP button was pressed, which shows that the physical safety constraints are working perfectly.

